

Class Activity: 10.07.2026

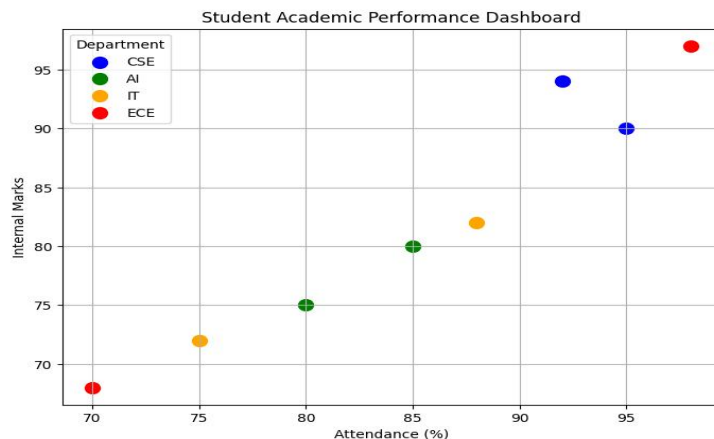
Scenario 1: Student Academic Performance Dashboard

Code:

```
install.packages("ggplot2")
library(ggplot2)
student_data <- data.frame(
  StudentID = c(101,102,103,104,105,106,107,108),
  Attendance = c(95,80,88,70,92,85,75,98),
  InternalMarks = c(90,75,82,68,94,80,72,97),
  Department = c("CSE", "AI", "IT", "ECE", "CSE", "AI", "IT", "ECE")
)

# Scatter Plot
ggplot(student_data,
  aes(x = Attendance,
      y = InternalMarks,
      color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks",
    color = "Department"
  ) +
  theme_minimal()
```

OUTPUT:



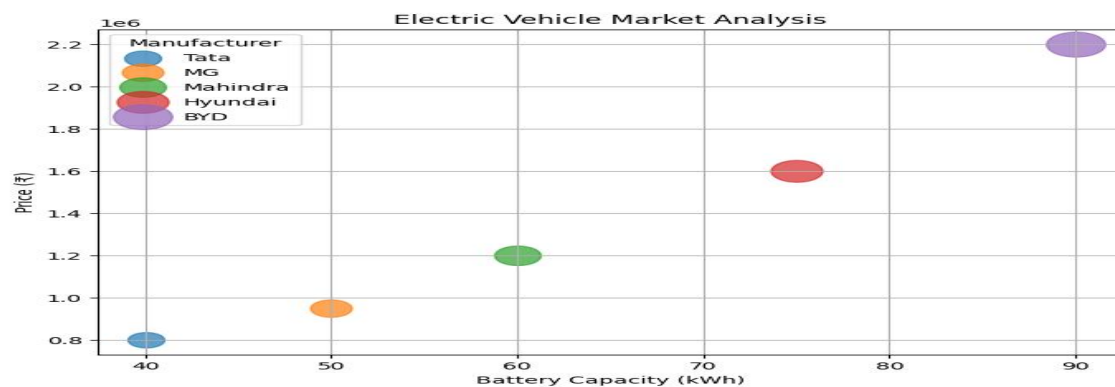
Scenario 2: Electric Vehicle Market Analysis

CODE:

```
install.packages("ggplot2")
library(ggplot2)
ev_data <- data.frame(
  VehicleModel = c("Model A", "Model B", "Model C", "Model D", "Model E"),
  BatteryCapacity = c(40, 50, 60, 75, 90),
  Price = c(800000, 950000, 1200000, 1600000, 2200000),
  DrivingRange = c(250, 320, 400, 500, 650),
  Manufacturer = c("Tata", "MG", "Mahindra", "Hyundai", "BYD")
)
```

```
)
ggplot(ev_data,
  aes(x = BatteryCapacity,
      y = Price,
      size = DrivingRange,
      color = Manufacturer)) +
geom_point(alpha = 0.8) +
labs(
  title = "Electric Vehicle Market Analysis",
  x = "Battery Capacity (kWh)",
  y = "Price (₹)",
  color = "Manufacturer",
  size = "Driving Range (km)"
) +
theme_minimal()
```

OUTPUT:



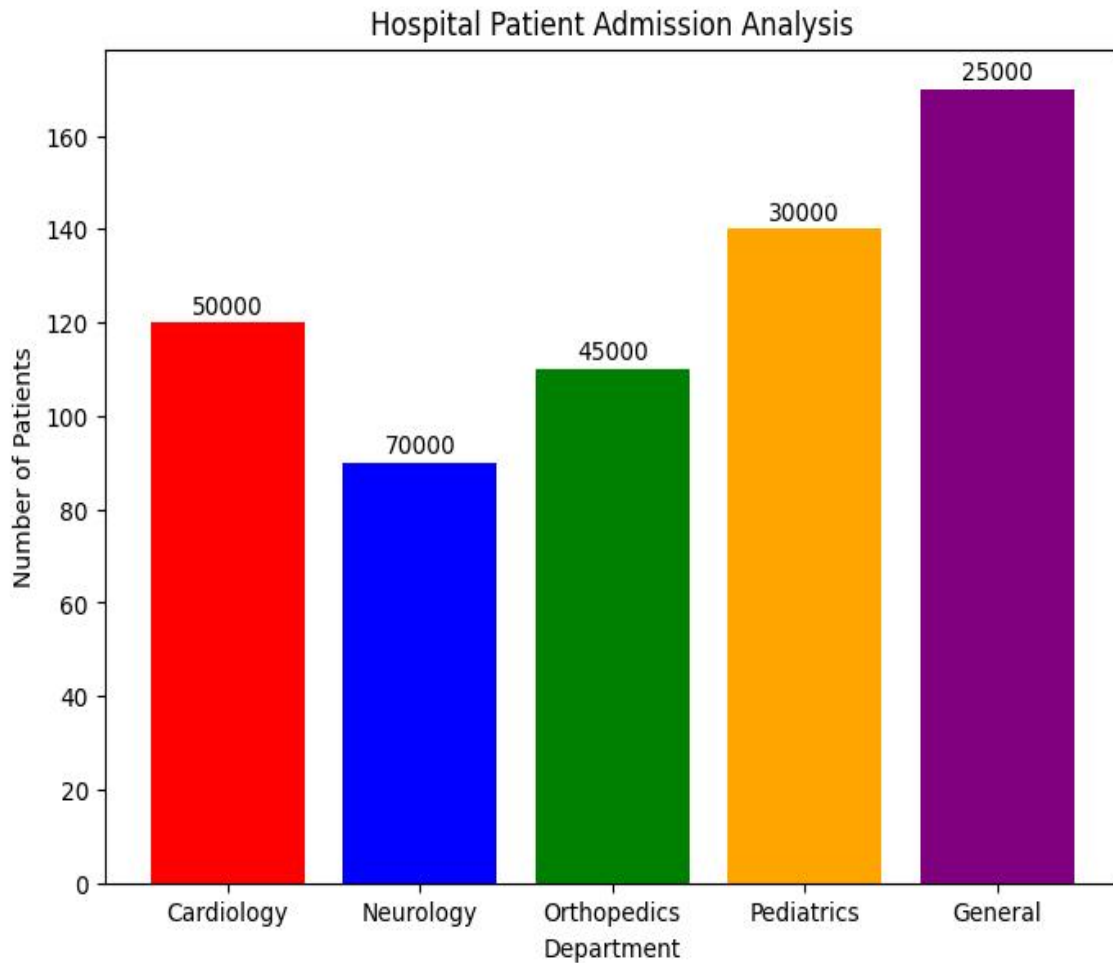
Scenario 3: Hospital Patient Admission Analysis

CODE:

```
install.packages("ggplot2")
library(ggplot2)
hospital_data <- data.frame(
  Department = c("Cardiology", "Neurology", "Orthopedics", "Pediatrics", "General"),
  Patients = c(120, 90, 110, 140, 170),
  TreatmentCost = c(50000, 70000, 45000, 30000, 25000)
)
ggplot(hospital_data,
  aes(x = Department,
      y = Patients,
      fill = Department)) +
geom_bar(stat = "identity") +
geom_text(aes(label = TreatmentCost),
  vjust = -0.4,
  size = 4) +
labs(
  title = "Hospital Patient Admission Analysis",
  x = "Department",
  y = "Number of Patients"
) +
```

theme_minimal()

OUTPUT:

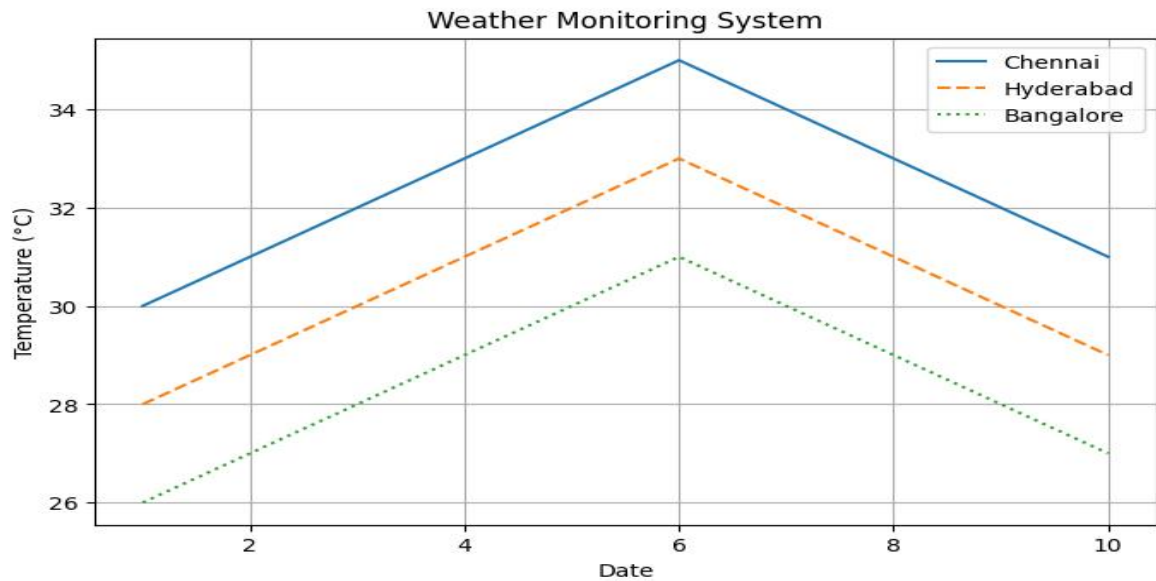


Scenario 4: Weather Monitoring System

CODE:

```
install.packages("ggplot2")
library(ggplot2)
weather <- data.frame(
  Date=rep(1:10,3),
  Temperature=c(30,31,32,33,34,35,34,33,32,31,
28,29,30,31,32,33,32,31,30,29,
26,27,28,29,30,31,30,29,28,27),
  City=rep(c("Chennai","Hyderabad","Bangalore"),each=10)
)
ggplot(weather,aes(x=Date,
  y=Temperature,
  color=City,
  linetype=City,
  group=City))+
  geom_line(size=1.2)+
  labs(title="Weather Monitoring System",
  x="Date",
  y="Temperature (°C)")+
  theme_minimal()
```

OUTPUT:

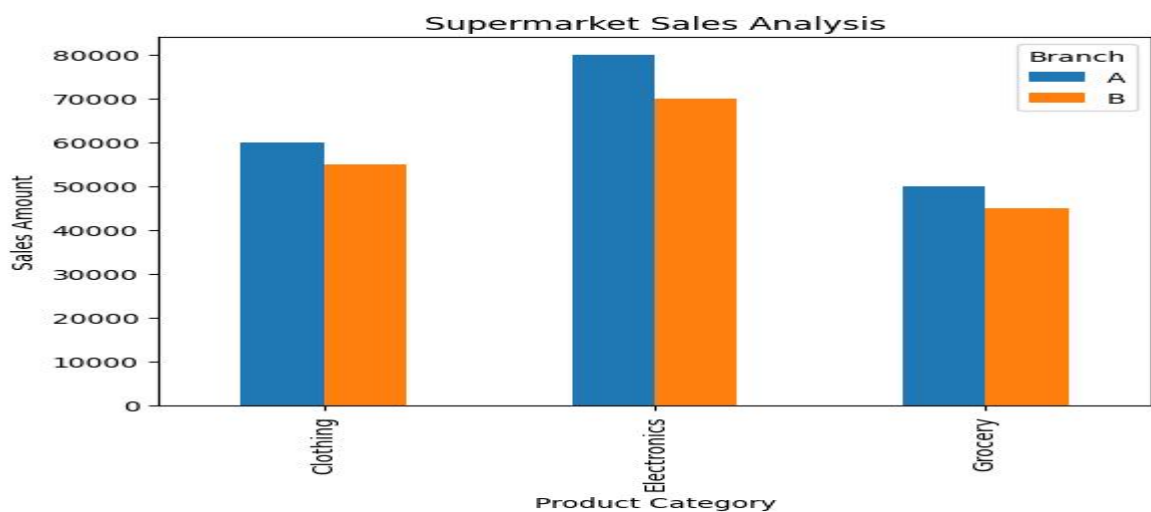


Scenario 5: Supermarket Sales Analysis

CODE:

```
library(ggplot2)
sales <- data.frame(
  Category=c("Grocery","Grocery","Electronics","Electronics","Clothing","Clothing"),
  Sales=c(50000,45000,80000,70000,60000,55000),
  Branch=c("A","B","A","B","A","B")
)
ggplot(sales,
  aes(x=Category,
    y=Sales,
    fill=Branch))+
  geom_bar(stat="identity",
    position="dodge")+
  labs(title="Supermarket Sales Analysis",
    x="Product Category",
    y="Sales Amount")+
  theme_minimal()
```

OUTPUT:

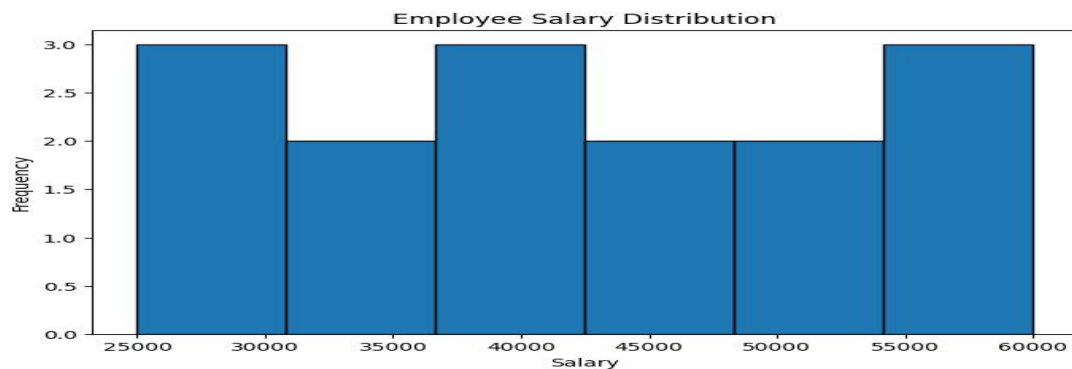


Scenario 6: Employee Salary Distribution

CODE:

```
library(ggplot2)
employee <- data.frame(
  EmployeeID=1:15,
  Salary=c(25000,28000,30000,32000,35000,
38000,40000,42000,45000,47000,
50000,52000,55000,58000,60000),
  Experience=c(1,2,2,3,4,5,5,6,7,8,9,10,11,12,13),
  Department=c("HR","IT","HR","Sales","IT","Sales",
"IT","HR","Finance","Finance","IT",
"Sales","Finance","HR","IT")
)
ggplot(employee,
  aes(x=Salary))+
  geom_histogram(binwidth=5000,
  fill="steelblue",
  color="black")+
  labs(title="Employee Salary Distribution",
  x="Salary",
  y="Frequency")+
  theme_minimal()
```

OUTPUT:

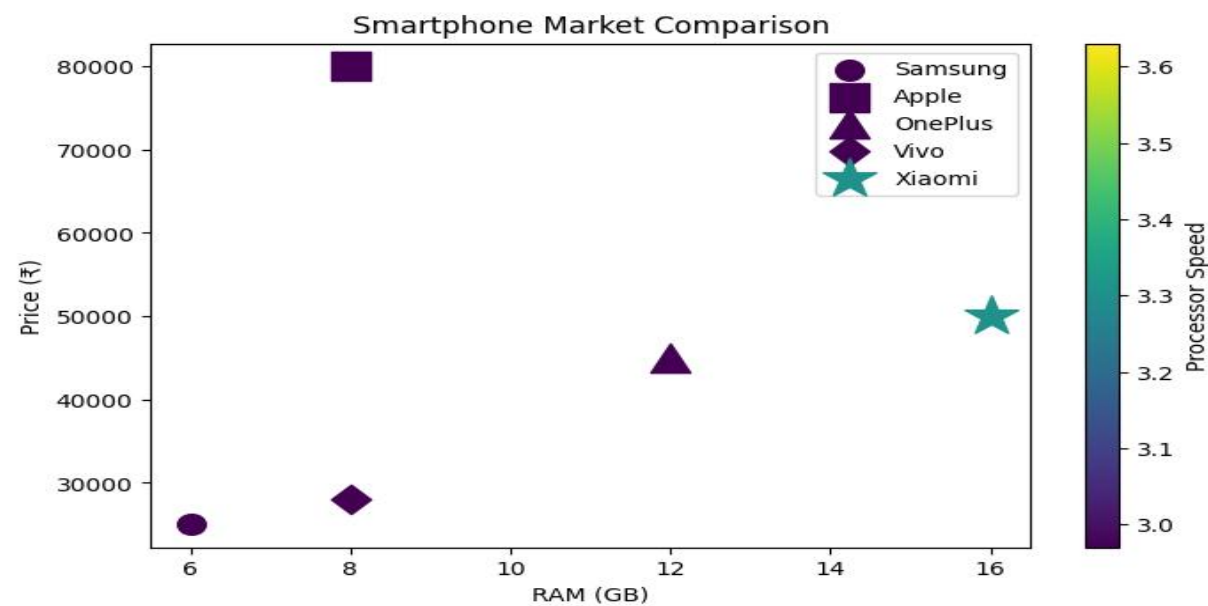


Scenario 7: Smartphone Market Comparison

CODE:

```
library(ggplot2)
phone <- data.frame(
  Brand=c("Samsung","Apple","OnePlus","Vivo","Xiaomi"),
  RAM=c(6,8,12,8,16),
  ProcessorSpeed=c(2.2,3.1,3.2,2.5,3.3),
  Price=c(25000,80000,45000,28000,50000),
  Storage=c(128,256,256,128,512)
)
ggplot(phone,
  aes(x=RAM,
  y=Price,
  shape=Brand,
  size=Storage,
  color=ProcessorSpeed))+
  geom_point()+
  labs(title="Smartphone Market Comparison",
  x="RAM (GB)",
  y="Price (₹)")+
  theme_minimal()
```

OUTPUT:

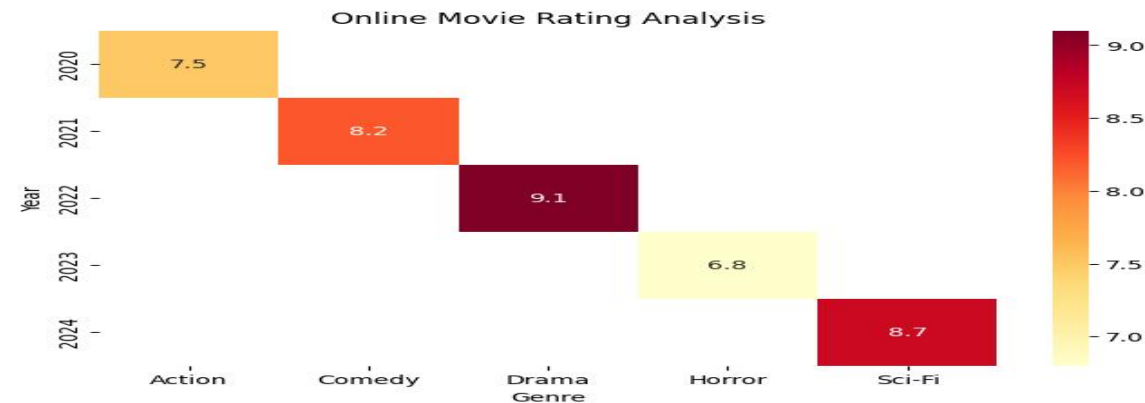


Scenario 8: Online Movie Rating Analysis

CODE:

```
library(ggplot2)
movie <- data.frame(
  Genre=c("Action","Comedy","Drama","Horror","Sci-Fi"),
  Rating=c(7.5,8.2,9.1,6.8,8.7),
  Reviews=c(5000,4000,7000,3000,6000),
  Year=c(2020,2021,2022,2023,2024)
)
ggplot(movie,
  aes(x=Genre,
    y=factor(Year),
    fill=Rating))+
  geom_tile(color="white")+
  scale_fill_gradient(low="yellow",
    high="red")+
  labs(title="Online Movie Rating Analysis",
    x="Genre",
    y="Release Year")+
  theme_minimal()
```

OUTPUT:

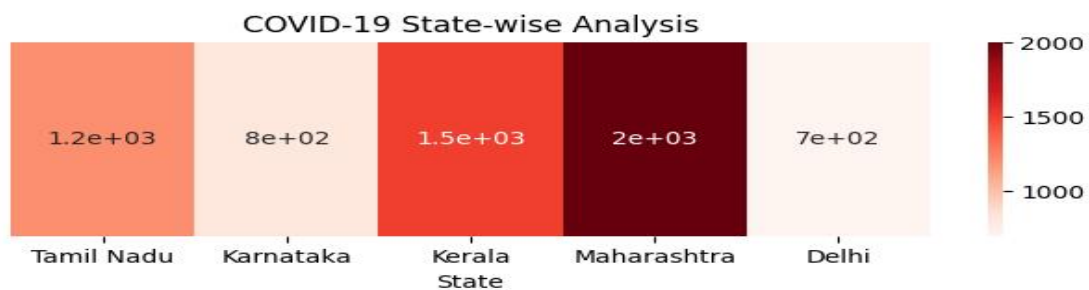


Scenario 9: COVID-19 State-wise Analysis

CODE:

```
library(ggplot2)
covid <- data.frame(
  State=c("Tamil Nadu","Karnataka","Kerala","Maharashtra","Delhi"),
  Cases=c(1200,800,1500,2000,700)
)
ggplot(covid,
  aes(x=State,
    y=1,
    fill=Cases))+
  geom_tile()+
  scale_fill_gradient(low="lightyellow",
    high="red")+
  labs(title="COVID-19 State-wise Analysis",
    x="State",
    y="")+
  theme_minimal()
```

OUTPUT:



Scenario 10: Iris Flower Classification

CODE:

```
library(ggplot2)
ggplot(iris,
  aes(x=Sepal.Length,
    y=Petal.Length,
    color=Species,
    shape=Species,
    size=Sepal.Width))+
  geom_point()+
  labs(title="Iris Flower Classification")+
  theme_minimal()
```

OUTPUT:

